PROJECT REPORT

Yudhishtra 2K26

Drone-Based Monitoring The Progress of Building Construction Projects

(SKYLINK APP – Construction Monitoring System)

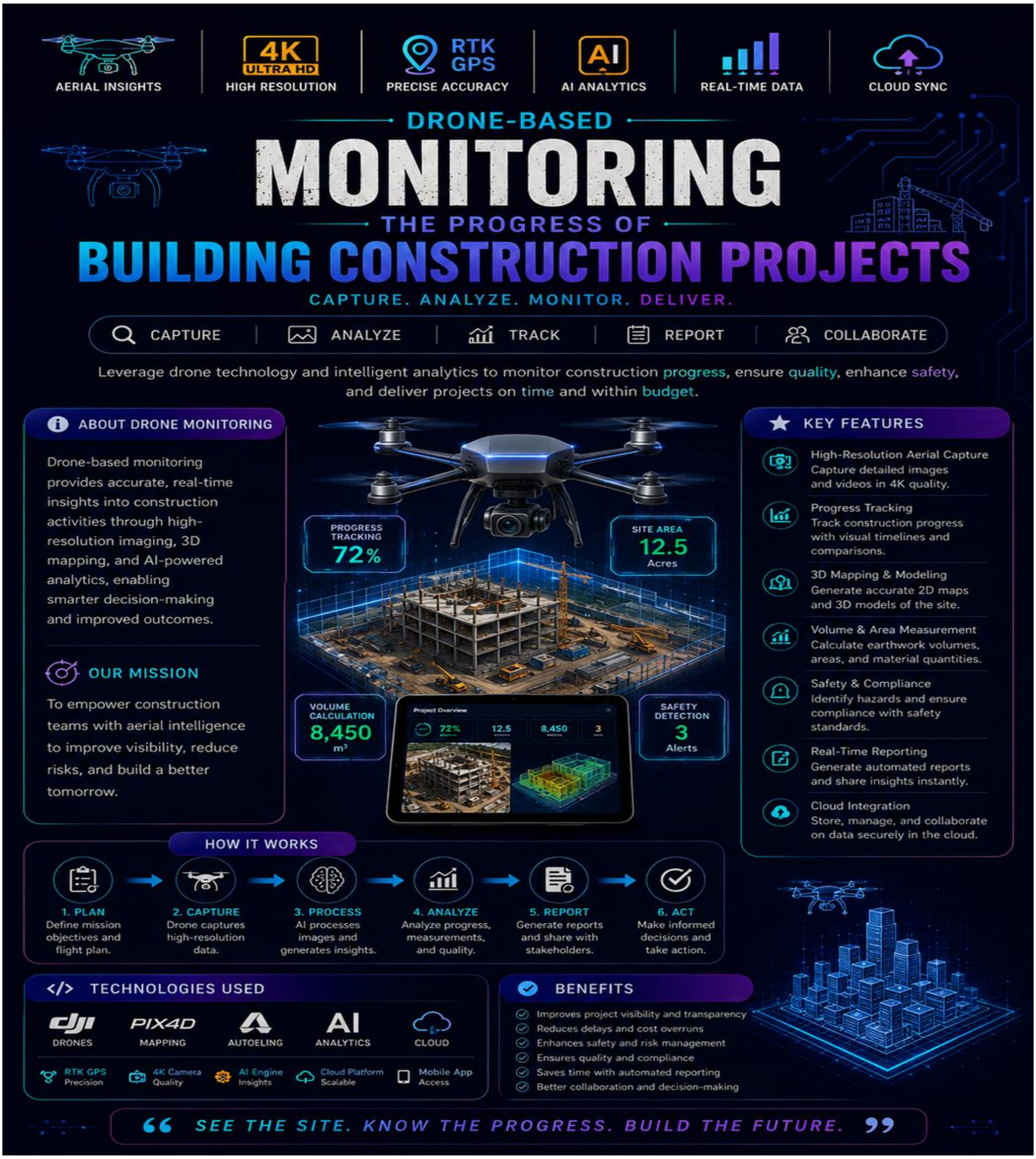

Figure 1: Project Poster – Yudhishtra Smart Monitoring System

Project Title	SkyLink App – Drone Based Construction Monitoring System
Project ID	C29RTCY117
Team Name	Tech Titans
Institution	Rathinam Technical Campus (Autonomous), Coimbatore
Department	Science & Humanities
Event	Project Yudhishtra 2K26
Live URL	https://sky-link-dashboard-irshath666.replit.app
Mentor	Ms. Anjitha NK
Date	June 2026

1. Team Members

S.No	Name	Role	Department
1	Lavanya A	Team Leader / Lead Developer	CSE-B
2	Irshath K	Frontend Developer	CSE-B
3	Gowtham S	UI/UX Designer	CSE-B
4	Gokul Ragav R	Drone System Operation	CSE-B
5	Jayanth J	Testing & QA	CSE-B
6	Harish R	Database & Auth	CSE-B

2. About the Project

The construction industry often faces challenges in monitoring project progress accurately and efficiently. Traditional site inspections are time-consuming, labour-intensive, and may not provide real-time information. Drone-based monitoring offers an innovative solution by using unmanned aerial vehicles (UAVs) equipped with high-resolution cameras and sensors to capture aerial images and videos of construction sites. These images are analysed to monitor progress, improve project management, and ensure timely completion of construction activities.

2.1 Problem Statement

Construction site monitoring is traditionally done through manual inspections, which are time-consuming, costly, and prone to human error. Project managers and stakeholders often lack real-time visibility into construction progress, leading to delays, budget overruns, and safety hazards. In large-scale building construction projects, tracking progress across multiple floors and zones simultaneously is practically impossible through ground-level inspection alone. There is a critical need for an automated, accurate, and cost-effective aerial monitoring system that can capture real-time site data, detect deviations from the plan, and provide actionable insights to improve construction efficiency and project management.

2.2 Objective

To develop a drone-based monitoring system that captures aerial images of construction sites

To analyse the progress of building construction

To provides real-time updates to project managers

To improve accuracy, reduce manual effort, enhance safety, and support better decision-making.

3. Key Features

3.1 User-Side Features

- High-resolution aerial image and video capture
- Real-time monitoring of construction progress
- AI-based image analysis for progress estimation
- Comparison of actual progress with project schedule
- Automatic generation of progress reports
- Cloud storage for easy access and data management
- Remote monitoring through web or mobile applications
- Improved safety by reducing manual site inspection

3.2 Admin Side Features

1. Dashboard
2. Project Management
3. Drone Management
4. Image & Video Management
5. Construction Progress Monitoring
6. AI Analysis
7. Report Generation
8. User Management

9. Notifications & Alerts

10. Site Safety Monitoring

4. Technology Stack

Category	Technology / Tool
Frontend	HTML5, CSS3, JavaScript (Glassmorphism UI)
Backend / Database	Python (Flask/Django), MySQL or Firebase Firestore
Hosting / Deployment	Render, Vercel (Frontend), Railway or PythonAnywhere (Backend)
Drone	DJI Drone / Any UAV with HD Camera
Image Processing	OpenCV
Email Service	SMTP (Gmail), EmailJS, SendGrid
IDE / AI Tools	Visual Studio Code, PyCharm, Google Colab, ChatGPT, GitHub Copilot
Version Control	Git & GitHub
AI Model	YOLOv8 / TensorFlow / PyTorch
Mapping	**Pix4D, DroneDeploy, OpenDroneMap**

5. How It Works

1. Plan the Mission

Define the construction site, flight path, altitude, and schedule for the drone inspection.

2. Drone Captures Images

The drone flies over the construction site and captures high-resolution aerial images and videos from multiple angles.

3. Upload & Process Data

The captured images are uploaded to the cloud/server where image processing and AI algorithms analyse the construction progress.

4. Analyse Construction Progress

AI compares the latest images with previous site images or the planned construction model to identify completed work, delays, and changes.

5. Generate Reports

The system creates daily, weekly, or monthly reports with progress charts, before-and-after images, and project status.

6. Monitor & Manage

The admin or project manager views the reports on the dashboard, receives alerts for delays or safety issues, and makes informed decisions.

6. Achievements & Recognition

Successfully developed a working prototype for drone-based construction monitoring.
Integrated drone image capture with AI-based progress analysis.
Designed an interactive admin dashboard for monitoring construction progress.
Automated progress report generation using aerial images.
Demonstrated real-time monitoring capabilities for construction projects.

7. Project Photos

7.1 Team at the Exhibition

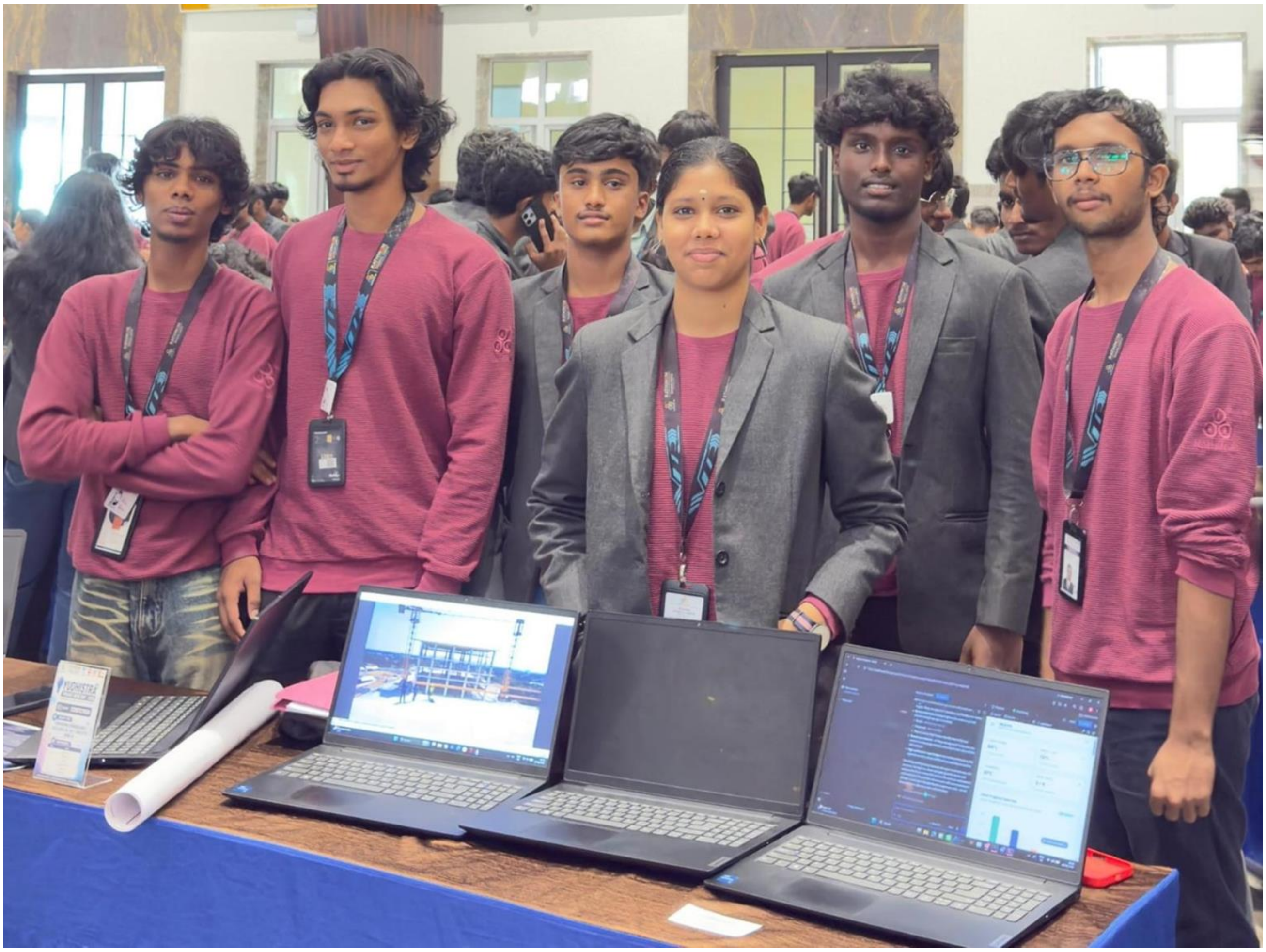

Figure 2: Team Tech Titans at the Yudhishtra 2K26 Exhibition – Rathinam Technical Campus

7.2 Project Display Board

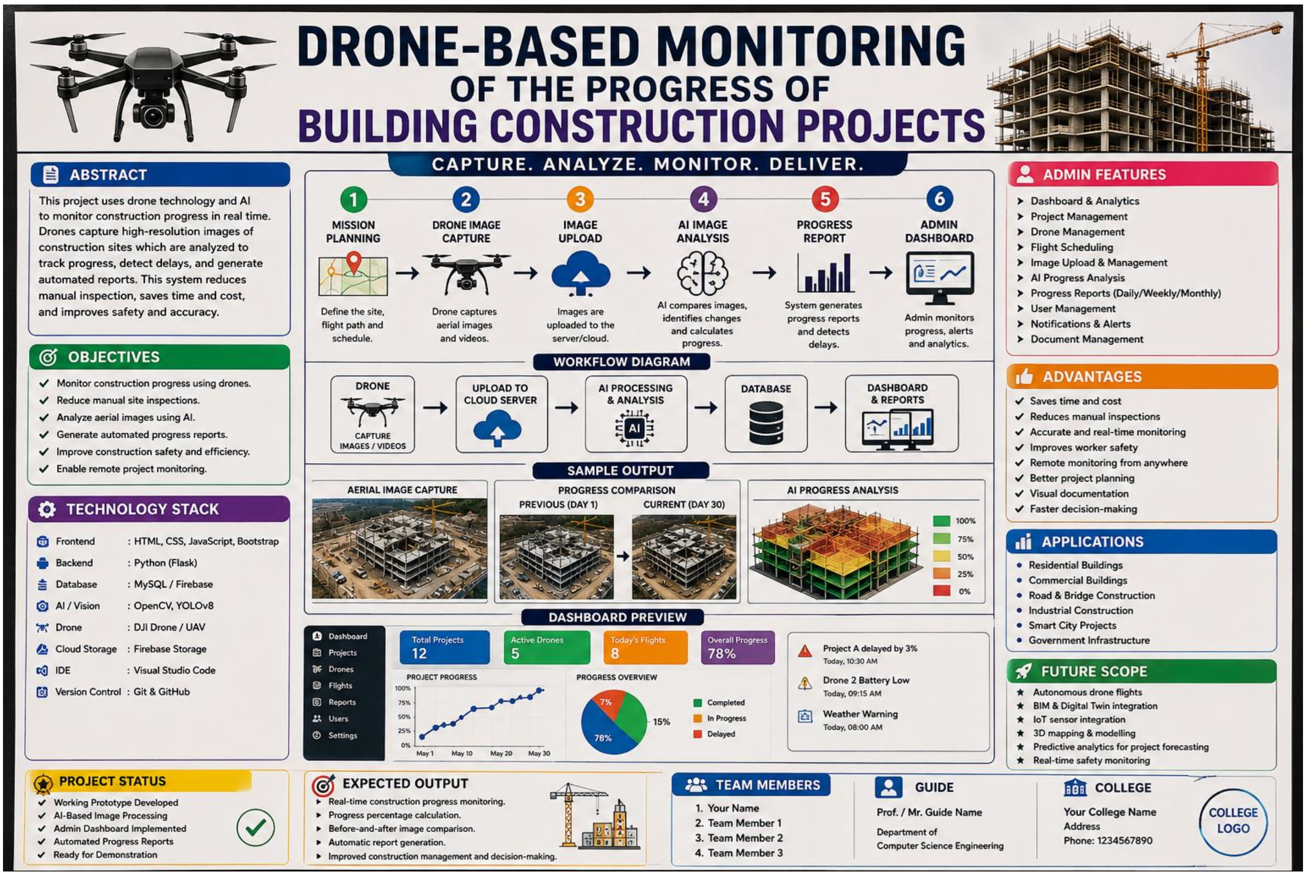

Figure 3: Display Board showing System Flow, Architecture, and UI Screens

8. Conclusion

Drone-Based Monitoring of Building Construction Projects is a modern and efficient solution for managing construction activities. By combining drone technology, image processing, and AI, the system enables accurate progress tracking, improves safety, reduces costs, and helps ensure projects are completed on time. This technology supports smarter decision-making and represents the future of digital construction management.

Future Scope

The system can be enhanced by integrating IoT sensors, GPS, BIM (Building Information Modeling), and advanced AI algorithms for automatic defect detection, material tracking, safety monitoring, and predictive project scheduling. Future versions may also support autonomous drone flights and digital twin technology for more accurate project visualization.

9. Declaration

We hereby declare that the project titled "Drone Based Construction Monitoring System" was independently designed, developed, and deployed by Team Tech Titans, first-year B.E. Computer Science and Engineering students of Rathinam Technical Campus (Autonomous), Coimbatore, under the guidance of our mentor Ms. Anjitha NK.

The project is an original work and has been submitted for Project Yudhishtra 2K26, organized by the Department of Science & Humanities.

Lavanya A (Team Leader)	Ms. Anjitha NK (Mentor)	Department of Science & Humanities